Medical Cost Linear Regression Model

1. Introduction

This project aims to build a machine learning model that predicts medical insurance costs based on patient demographic and health-related features. The model uses historical data to understand the relationship between personal attributes such as age, BMI, smoking status, and medical charges.

2. Problem Definition

The goal of this project is to predict continuous medical insurance charges based on multiple input features. Since the target variable is numerical and continuous, this is a regression problem.

3.Dataset Description

The dataset contains 6 input features and 1 target variable:

- Age
- Sex
- BMI
- Children
- Smoker
- Region
- Charges (Target variable)

The dataset represents real-world medical insurance records used to estimate patient healthcare costs.

4. Features Explanation

- **Age:** Age of the insured person
- **Sex:** Gender of the individual
- **BMI:** Body Mass Index indicating body fat level
- **Children:** Number of dependents covered
- **Smoker:** Smoking status (yes/no)
- **Region:** Residential area in the US
- **Charges:** Annual medical insurance cost (target variable)

Note : Label Encoding was applied to convert categorical (text-based) variables into numerical values because machine learning algorithms cannot process non-numeric data. This step was necessary since the dataset contained both numerical and textual features. Each category was assigned a unique integer value to make the data suitable for model training.

5.Methodology

1. Data loading and inspection
2. Data cleaning and preprocessing
3. Encoding categorical variables
4. Splitting dataset into training and testing sets
5. Training a Linear Regression model
6. Evaluating model performance

6.Model Selection

Linear Regression was chosen because it is suitable for predicting continuous numerical values and helps model the linear relationship between independent variables and medical charges.

7. Model Evaluation

Mean Absolute Error (MAE)

Root Mean Square Error (RMSE)

Coefficient of Determination (R^2)

The model was evaluated using the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2). The results indicate how well the model fits the data and its ability to predict unseen samples.

8.Model Interpretation

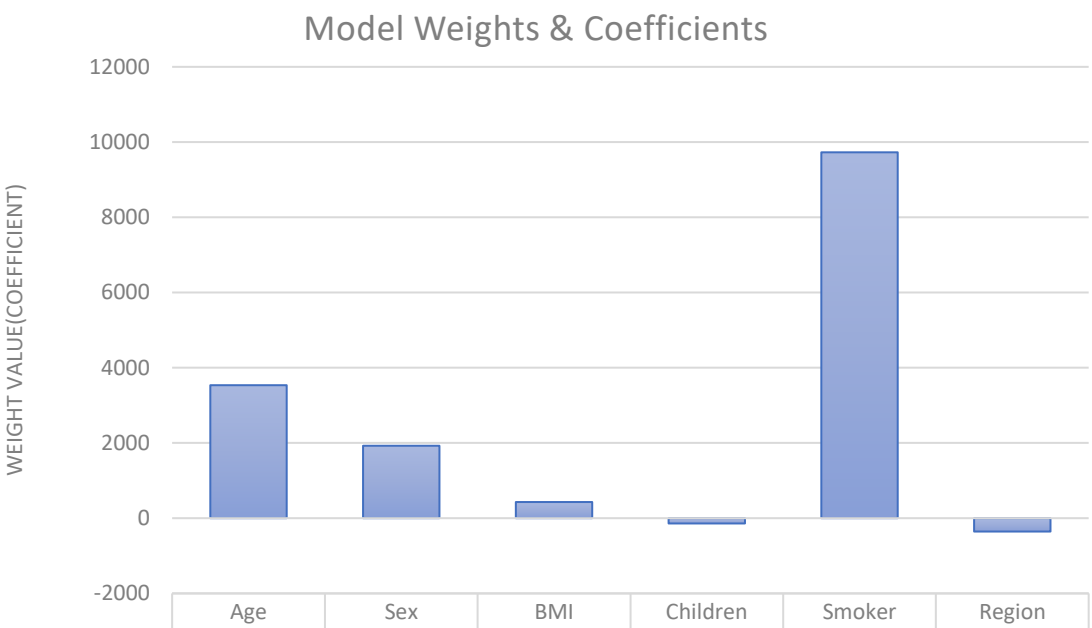
The model shows that smoking status and BMI have the strongest impact on medical costs. Higher BMI and smoking behavior significantly increase predicted charges, while other features contribute moderately.

9.Prediction Example

The trained model was tested on a new unseen sample to predict medical costs, demonstrating its ability to generalize beyond training data

Conclusion

This project demonstrates how linear regression can be used to predict medical insurance costs based on patient data. The model provides useful insights for insurance pricing strategies and highlights the importance of health-related factors in cost prediction. Future improvements may include using advanced models such as Random Forest or Gradient Boosting to improve accuracy.



	Age	Sex	BMI	Children	Smoker	Region
Model Weights	3535.61	1925.91	429.49	-140.54	9727.71	-354.97

The work of the two female engineers:
Ranim Tayseer Alqrenawi & Saja Nabeel Nabhan